Talank Baral

Ph.D. Student in Computer Science
Department of Computer Science, George Mason University • Falls Church, VA 22046, USA
tbaral@gmu.edu talank.com.np github.com/talank in/tlnkb

Summary

Software engineer and Ph.D. student in Computer Science (expected May 2027) focused on testing, continuous integration, and build reliability. Over two years in industry building and maintaining automated end-to-end, API, and GUI test suites and Docker-based CI pipelines for ownCloud, an open-source file-sync and collaboration platform, followed by four years of research building tools that make CI builds faster and tests more reliable.

Education

2022–Present **Ph.D. in Computer Science** — Fairfax, VA, USA
George Mason University
GPA: 3.98/4.00 • Advisor: Wing Lam; expected graduation May 2027

2015–2019 **B.E. in Software Engineering** — Pokhara, Nepal
Gandaki College of Engineering and Science, Pokhara University
GPA: 3.67/4.00

Experience

Aug 2022–Present **Graduate Research/Teaching Assistant** — Fairfax, VA, USA
Department of Computer Science, George Mason University

- Reproduced timing-dependent flaky test failures with a neurosymbolic LLM pipeline (TDRepro; ASE ’26 Distinguished Paper)
- Detected and prevented unnecessary content generation in CI builds (OptCD; ASE ’23 and ICSE ’25 Demo)
- Studied the test coverage of pull requests authored by AI coding agents (ICSME ’26)
- Developed techniques for efficient test orders in regression testing

Jan 2020–Aug 2022 **Software Developer** — Pokhara, Nepal
JankariTech

- Built and maintained BDD (Gherkin) acceptance test suites for ownCloud, an open-source file-sync and collaboration platform: web UI (Nightwatch.js), server WebDAV and sharing APIs (Behat), and the desktop sync client (Squish GUI tests); 100+ merged pull requests
- Made CI failures diagnosable: HTTP and server logs, desktop screenshots on test failure, stack traces for crashed applications under test, and a parser for GUI test logs
- Maintained Drone CI pipelines, including nightly cron runs with chat notifications and Dockerized test middleware, and kept integration test suites in sync across ownCloud server, oCIS, and Reva
- Refactored tests to the Page Object Model and fixed intermittent failures and tests that passed when they should have failed
- Set up test automation, CI pipelines, and acceptance tests for client projects

Selected Projects

2023–2025 **OptCD** [Code] [Demo]
Python, Bash, GitHub Actions, Maven, Gemini API

- Instruments a GitHub Actions workflow to find directories generated but never accessed, traces

each to the responsible Maven plugin and step, and uses an LLM (Gemini) to propose a fixed workflow that it pushes for validation in CI
- Built the evaluation pipeline: mined Java repositories through the GitHub API, filtered for Maven and GitHub Actions projects, and aggregated file-access logs from 600+ CI job runs into per-job and per-project results
- Findings led to 25 pull requests to open-source projects that disable unnecessary generation

2016–2021 **Chamena Griha** [Code]

PHP, MySQL, JavaScript, Nightwatch.js, Cucumber, Selenium, Docker, Travis CI

- Food-ordering web app for college canteens with separate student, canteen, and college-admin roles
- Added Gherkin acceptance tests (Nightwatch.js + Cucumber) and a Travis CI pipeline that provisions MySQL, seeds the schema, and runs Selenium in Docker

2018–2019 **Gas Station Management System** [Code] [Demo]

Java, JavaFX, Apache Derby, Maven

- Desktop point-of-sale and accounting app for a fuel station built with JavaFX and an embedded Apache Derby database
- Covers instant and credit billing, credit-company accounts, fuel stock and tank refills, daily closing reports, and receipt printing

Technical Skills

Languages	Python, Java, JavaScript, Bash, SQL, PHP, C, C++
Testing	JUnit, Selenium, Cypress, Nightwatch.js, Jest, Puppeteer, Behat, Squish, Cucumber (BDD/Gherkin)
CI/CD	GitHub Actions, GitLab CI, Travis CI, Drone, Docker, Maven, Git
Data & APIs	MySQL, Apache Derby, GitHub REST API, Gemini API

Open-Source Contributions

- **International Dataset of Flaky Tests (IDoFT)**: Detected and fixed flaky tests in open-source Java projects
- **pact-cplusplus**: Fixed compilation issues in version 0.1.0

Publications

Conference Papers

[C3] ASE '26

Shanto Rahman, **Talank Baral**, August Shi, and Wing Lam. **TDRepro: A Neurosymbolic Approach to Reproducing Timing-Dependent Flaky Test Failures**. *41st IEEE/ACM International Conference on Automated Software Engineering (ASE 2026)*, pp. 675–686, Munich, Germany, 2026.

ACM SIGSOFT Distinguished Paper Award

Acceptance rate: 8.82% direct acceptance [DOI] [PDF]

[C2] ICSME '26

Atish Kumar Dipongkor, **Talank Baral**, Wing Lam, and Kevin Moran. **Test Coverage Analysis of Agentic Pull Requests**. *42nd International Conference on Software Maintenance and Evolution (ICSME 2026)*, Vision and Emerging Results Track, Benevento, Italy, 2026.

Acceptance rate: 39% [arXiv] [Code] [Artifact]

[C1] ASE '23 — **Talank Baral**, Shanto Rahman, Bala Naren Chanumolu, Basak Balci, Tuna Tuncer, August Shi, and Wing Lam. **Optimizing Continuous Development By Detecting and Preventing Unnecessary Content Generation**. *38th IEEE/ACM International Conference on Automated Software Engineering (ASE 2023)*, pp. 901–913, Kirchberg, Luxembourg, 2023.
Acceptance rate: 21% (134/630) [PDF] [Slides]

Tool Demonstrations

[D1] ICSE '25 Demo — **Talank Baral**, Emirhan Oğul, Shanto Rahman, August Shi, and Wing Lam. **OptCD: Optimizing Continuous Development**. *47th IEEE/ACM International Conference on Software Engineering (ICSE 2025)*, Demonstrations Track, pp. 45–48, Ottawa, ON, Canada, 2025.
[PDF] [Slides]

Under Review

[U1] **Talank Baral**, Abdelrahman Baz, Atish Kumar Dipongkor, August Shi, Kevin Moran, and Wing Lam. **Using Efficient Test Orders for Regression Testing**. *Under review*.

Honors & Awards

Oct 2026 — **ACM SIGSOFT Distinguished Paper Award**, ASE '26
TDRepro: A Neurosymbolic Approach to Reproducing Timing-Dependent Flaky Test Failures

Aug 2023 — **SIGSOFT CAPS Travel Grant**, ACM SIGSOFT
Awarded $1,200 to attend ASE 2023 (declined due to visa constraints)

May 2023 — **Summer Research Award**, Department of Computer Science, George Mason University

2015–2019 — **Undergraduate Academic Excellence Award**, Gandaki College of Engineering and Science, Pokhara University
Merit-based award

Feb 2018 — **First Place, Project Expo**, 7th GCES IT Expo, Pokhara University

Jun 2012 — **National Talent Identification Award (First Place Nationwide)**, Government of Nepal
For designing a seed plantation machine

Teaching

Fall 2026	**SWE 637 / SWE 437**: Software Testing	*Graduate Teaching Assistant*
Spring 2026	**CS 691**: Synergies of LLMs and Software Engineering	*Graduate Teaching Assistant*
Fall 2022	**CS 262**: Introduction to Low-Level Programming	*Graduate Teaching Assistant*

Invited & Conference Talks

May 2025 — *OptCD: Optimizing Continuous Development*
ICSE 2025 (Demonstrations Track) • Conference Talk • Ottawa, ON, Canada

Jan 2024 — *Optimizing Continuous Development By Detecting and Preventing Unnecessary Content Generation*
ESSEC 2024 • Colloquium Talk • William & Mary, Williamsburg, VA, USA

Jul 2023 — *Optimizing Continuous Development*
Microsoft, Redmond Campus • Industry Talk • Redmond, WA, USA

Jul 2023 — *Optimizing Continuous Development and Predicting Test Information in Software Development*
ISSTA 2023 • Doctoral Symposium • Seattle, WA, USA